Group-Structured Attributions for Explainable Quantum Machine Learning

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***Abstract*—Explainable quantum machine learning needs methods that can identify not only important individual gates, but also important functional parts of a quantum circuit. This thesis adapts Owen values to quantum-circuit explanations by treating active gates as players, predefined circuit modules as groups, and reduced circuits as coalition evaluations. The method is evaluated on a controlled benchmark of three-qubit circuits using quantum-resource value functions, and on two quantum machine learning case studies using classification accuracy. The benchmark results show that Owen values produce meaningful group-level explanations of quantum resources: the dominant groups change according to the value function being explained and align with the expected functional roles of the circuit modules. Group ablation further supports behavioural faithfulness, since removing highly ranked groups generally causes larger changes in the target value than removing low-ranked groups. A Monte Carlo estimator preserves the main Owen-value rankings while reducing exact enumeration cost. In the quantum machine learning case studies, Owen values retain the useful gate-level patterns of Shapley-based explanations while adding a clearer module-level interpretation. These results show that Owen values provide a useful complementary explanation framework for group-structured quantum circuits.**

I. Introduction

Quantum Machine Learning (QML) is a fast-growing field at the intersection of machine learning and quantum computing [1], [2]. Many QML approaches use parameterized quantum circuits as part of machine-learning models [3]. As QML circuits become more complex, it becomes harder to understand which parts of the circuit are actually important. This motivates explainable quantum machine learning (XQML): methods that help us inspect quantum models and understand why they behave in a certain way [4], [5].

A recent approach called Shapley values for quantum circuit explanations (SVQX) explains quantum circuits by treating gates as players in a cooperative game [5]. Each gate receives a Shapley value, which measures how much that gate contributes to a chosen value function, such as classification accuracy. This builds on the broader use of Shapley values in explainable artificial intelligence [6].

However, quantum circuits are often not best understood as lists of separate gates. They are usually designed from larger functional parts, where different gates or groups of gates contribute to different circuit functions [7]. For example, some parts of a circuit may mainly contribute to one quantum resource, while others may contribute to another, and these resources can interact within the same circuit [8]. A purely gate-level explanation may therefore be too detailed to show which larger circuit mechanism is responsible for a result.

This thesis investigates whether Owen values can address this limitation. Owen values extend Shapley values to games where players are divided into predefined groups [9], [10]. To the best of our knowledge, Owen values have not previously been applied to quantum-circuit explanation. We therefore propose an Owen-value framework that produces attributions at both the gate level and the functional-group level, evaluated on a controlled benchmark and two QML case studies.

These gaps motivate the following research questions:

RQ1: *How can Owen values be adapted and computed, exactly and approximately, to explain the importance of individual gates and functional groups in group-structured quantum circuits?*

RQ2: *To what extent do Owen values produce meaningful and behaviourally faithful group-wise explanations on a controlled quantum-circuit benchmark?*

RQ3: *How do Owen-value explanations compare with standard gate-level Shapley-value explanations in QML case studies, and what additional module-level interpretations do Owen values provide?*

II. Background

Shapley and Owen values are cooperative-game attribution methods that can explain quantum circuits by treating gates as players, modules as groups, and circuit properties or accuracy as coalition values.

A. Shapley and Owen Values

A cooperative game is defined by a set of players $N = \{1, \dots, n\}$ and a value function $v : 2^N \to \mathbb{R}$. The value $v(S)$ describes the score obtained by a coalition $S \subseteq N$. The Shapley value of player i is its average marginal contribution over all coalitions not containing i [11]:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} \left[v(S \cup \{i\}) - v(S) \right]. \tag{1}$$

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The Shapley value is used in explainable artificial intelligence to divide a model output among features or model components [6]. In quantum-circuit explanations, the players can be gates and $v(S)$ can measure the output of the reduced circuit that keeps only the gates in S [5].

Unlike Shapley values, Owen values incorporate a predefined group structure over the players [9], [10]. If $\mathcal{P} = \{G_1, \ldots, G_m\}$ is a partition of N into groups, then $G(i)$ denotes the group containing player i. For player i, an Owen context is defined by two coalitions: an outer coalition $R \subseteq \mathcal{P} \setminus \{G(i)\}$ of groups, and an inner coalition $T \subseteq G(i) \setminus \{i\}$ of players inside the group of i. The union of the groups in R is

$$U_R = \bigcup_{G \in R} G. \tag{2}$$

The marginal contribution of player i in this Owen context is

$$\Delta_i(R, T) = v(U_R \cup T \cup \{i\}) - v(U_R \cup T)\,. \tag{3}$$

The Owen value of player i is the weighted average of these marginal contributions:

$$\psi_i(v, \mathcal{P}) = \sum_{R \subseteq \mathcal{P} \setminus \{G(i)\}} \sum_{T \subseteq G(i) \setminus \{i\}} \alpha(R, T) \Delta_i(R, T), \tag{4}$$

where

$$\alpha(R, T) = \frac{|R|!(m - |R| - 1)!}{m!} \frac{|T|!(|G(i)| - |T| - 1)!}{|G(i)|!}. \tag{5}$$

Owen values thus measure a player's importance while respecting the chosen group structure. In this thesis, group-level importance is obtained by summing the Owen values of all active gates in the same group:

$$\Psi_G = \sum_{i \in G} \psi_i(v, \mathcal{P}). \tag{6}$$

B. Owen Values Under Uncertainty

The Owen value formula is deterministic once the value function is known. However, in quantum-circuit experiments, coalition values are obtained by running or simulating reduced circuits and may therefore be uncertain. This uncertainty can arise from finite measurement shots, hardware noise, or one-shot model predictions. In such cases, the observed coalition value contains random evaluation noise, following the uncertainty view used in SVQX [5].

A simple way to reduce this uncertainty is to evaluate the same coalition several times and average the results:

$$\bar{v}_K(S) = \frac{1}{K} \sum_{k=1}^{K} \tilde{v}^{(k)}(S). \tag{7}$$

Here, $\tilde{v}^{(k)}(S)$ is the observed value of coalition S in the k-th evaluation, and K is the number of repeated evaluations of the same coalition. Larger K usually gives a more stable coalition value, but also increases runtime. Using these averaged values, the marginal contribution of gate i in an Owen context is estimated as

$$\widehat{\Delta}_{i,K}(R, T) = \bar{v}_K(U_R \cup T \cup \{i\}) - \bar{v}_K(U_R \cup T), \tag{8}$$

where R is a coalition of groups, T is a coalition inside the group of gate i, and U_R is the union of the groups in R.

C. Quantum Circuit Explanations

In quantum-circuit explanations, active gates are treated as players in a cooperative game, while passive gates are kept fixed in every coalition evaluation. For a circuit C with active-gate set N, each coalition $S \subseteq N$ defines a reduced circuit C_S that contains all passive gates and only the active gates in S, in the original gate order. For a circuit property f, the value function is

$$v_f(S) = f(C_S). \tag{9}$$

Using Owen values with this reduced-circuit construction gives both gate-level attributions and group-level attributions for the chosen property.

D. Value Functions

A value function defines what property or behaviour is being explained. In this thesis, the 35-circuit benchmark uses property-based value functions, where the score is a physical property of the output quantum state. The QML case studies use a task-based value function, where the score is classification accuracy.

For the property-based experiments, each reduced circuit C_S is applied to the standard all-zero input state. For a circuit on q qubits, this gives the output state

$$|\psi_S\rangle = C_S|0\rangle^{\otimes q}. \tag{10}$$

The first property is magic, measured by stabilizer Rényi entropy (SRE) [12], [13]. Magic is a quantum resource associated with states outside the stabilizer/Clifford setting, enabling universal quantum computation when combined with Clifford operations [14]. For a q-qubit pure state $|\psi\rangle$, the stabilizer 2-Rényi entropy used in this thesis is

$$M_2(|\psi\rangle) = -\log\left(\frac{1}{2^q} \sum_{P \in \mathcal{P}_q} \langle\psi|P|\psi\rangle^4\right), \tag{11}$$

where $\mathcal{P}_q$ is the set of q-qubit Pauli strings. The logarithm is the natural logarithm. Stabilizer states have zero SRE, while larger values indicate more magic. The magic value function is therefore

$$v_{\text{magic}}(S) = M_2(|\psi_S\rangle). \tag{12}$$

The second property is entanglement. This thesis uses the Meyer–Wallach global entanglement measure [15], following Brennen's formulation for pure multi-qubit states [16]. For a q-qubit pure state $|\psi\rangle$, ρ_k is the reduced density matrix of qubit k. The Meyer–Wallach measure is

$$Q(|\psi\rangle) = 2\left(1 - \frac{1}{q} \sum_{k=1}^{q} \text{Tr}(\rho_k^2)\right). \tag{13}$$

The entanglement value function is therefore

$$v_{\text{ent}}(S) = Q(|\psi_S\rangle). \tag{14}$$

In this thesis, the property-based value functions are deterministic because they are computed using noiseless statevector simulation. For every reduced circuit C_S, the full output state $|\psi_S\rangle$ is obtained directly instead of being estimated from finite measurement shots.

For the QML case studies, the value function is classification accuracy. For a test set $\mathcal{D}_{\text{test}} = \{(x_j, y_j)\}_{j=1}^{d_{\text{test}}}$, the accuracy value function is

$$v_{\text{acc}}(S) = \frac{1}{d_{\text{test}}} \sum_{j=1}^{d_{\text{test}}} \mathbf{1}\left[\hat{y}_j^{(S)} = y_j\right], \tag{15}$$

where $\hat{y}_j^{(S)}$ is the prediction produced by the reduced model for input x_j.

Thus, the Owen-value framework is not tied to one specific value function: SRE and Meyer–Wallach entanglement are used to explain quantum resources, while accuracy is used to explain QML model performance [5].

III. Methods

This thesis extends the SVQX reduced-circuit idea with Owen values. The Owen-value framework is applied to a controlled 35-circuit magic–entanglement benchmark, in which active gates form the player set, circuit modules define the coalition structure, and property-based value functions define what is being explained.

A. Group-Structured Circuit Explanations

For each circuit, gates are divided into active and passive gates. Passive gates are kept in every reduced circuit but do not receive attributions. For a coalition S of active gates, the reduced circuit C_S contains all passive gates and only the active gates in S, preserving the original gate order. Active gates are further divided into groups, with group-level scores obtained via Eq. (6), producing both gate-level and group-level explanations.

B. Coalition Structure

The main coalition structure used in this thesis is a resource-based E/M/X partition. Active gates are divided into three groups:

- E: entanglement-related gates, mainly CNOT/CZ gates;
- M: local magic-related gates, mainly non-Clifford or parameterized single-qubit gates;
- X: mixed gates, mainly non-Clifford or parameterized gates that are embedded in, or directly adjacent to, entangling motifs.

Gates outside these active roles are treated as passive. The partition is manually designed from physical intuition, with groups corresponding to entanglement generation, local magic generation, and mixed local–entangling effects.

The E/M/X partition is used for the 35-circuit benchmark and as the main resource-based partition in the QML case studies. Because the circuits have different structures, the exact gate assignments are adapted to each circuit. The full benchmark partitions with the circuit structures are reported in Appendix A-A.

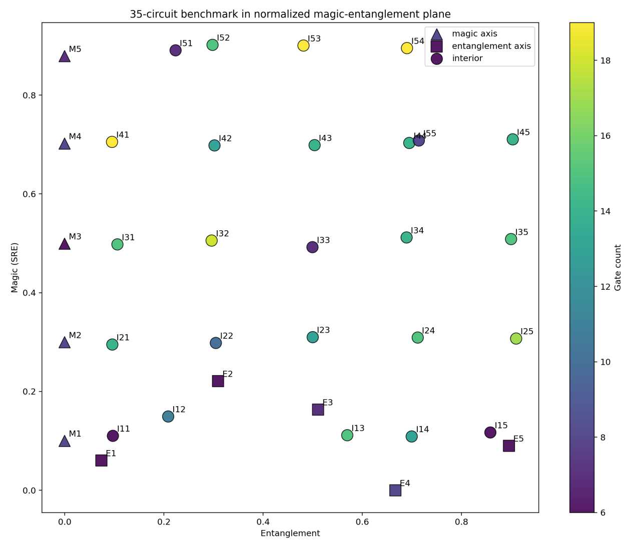

Fig. 1. Controlled benchmark of 35 three-qubit circuits in the normalized magic–entanglement plane.

C. Controlled Benchmark Construction

A controlled benchmark is used to assess whether Owen values can separate magic-related and entanglement-related circuit contributions. A pool of 10,000 random three-qubit circuits was generated as a large but computationally manageable search space for selecting circuits with different magic and entanglement levels. The circuit-generation and SRE-labelling code was adapted from the repository accompanying Lipardi et al. [13]. For each circuit, the number of gates was sampled uniformly from 6 to 19. One of three gate families was selected with equal probability: rotations with CNOT gates, Clifford-like gates with rotations and CNOT gates, or Clifford+T gates. Rotation angles were sampled uniformly from $[0, 2\pi)$.

Each circuit was labelled using the two property-based value functions from Section II-D: SRE for magic and the Meyer–Wallach measure for entanglement. Both values were normalized over the full pool. From this labelled pool, 35 circuits were selected to cover the normalized magic–entanglement plane, as shown in Fig. 1. The benchmark was designed to contain five magic-axis circuits with low entanglement and increasing magic, five entanglement-axis circuits with low magic and increasing entanglement, and 25 interior circuits with mixed values. Because the benchmark is selected from a finite random pool of small three-qubit circuits, the axes are approximate rather than perfectly ideal. In particular, the entanglement-axis circuits were chosen to have low magic while increasing in entanglement, but they do not all have exactly zero SRE. This reflects the fact that, for the available gate families, circuit depths and number of qubits, magic and entanglement cannot always be controlled independently.

The 15 shortest circuits are used for exact Owen computation, where all Owen contexts can still be enumerated. The full set of 35 circuits is used for the sampled Owen study.

All benchmark value functions are computed with noiseless statevector simulation. Thus, the SRE and Meyer–Wallach values are deterministic and are not estimated from measurement shots.

D. Exact and Sampled Owen Computation

Owen values can be computed exactly when the number of active gates and groups is small enough. Exact computation means that all Owen contexts are evaluated: for each active gate, all outer coalitions of groups and all inner coalitions inside the gate's own group are enumerated. In this thesis, exact Owen computation is used for the 15 short benchmark circuits and for the QML case studies.

For larger circuits, exact enumeration becomes expensive. A Monte Carlo Owen estimator is therefore introduced for the full 35-circuit benchmark, adapting the coalitional-value estimator of Kotsiopoulos et al. [17] to sample only Owen contexts (R, T), since the value function is evaluated directly for each context. For gate i, L contexts (R_ℓ, T_ℓ) are sampled with probabilities proportional to the Owen weights $\alpha(R, T)$ in Eq. (5), giving

$$\widehat{\psi}_i = \frac{1}{L}\sum_{\ell=1}^{L}\left[v(U_{R_\ell} \cup T_\ell \cup \{i\}) - v(U_{R_\ell} \cup T_\ell)\right], \qquad (16)$$

where U_{R_ℓ} is the union of the groups in R_ℓ. The weights do not appear explicitly in Eq. (16) because they are used as the sampling probabilities.

For stochastic value functions, each coalition value in Eq. (16) is replaced by its averaged estimate $\bar{v}_K$ from Eq. (7). For deterministic value functions $K = 1$ is sufficient.

The estimator is validated on the exact 15-circuit subset using sampling fractions 0.30, 0.50, 0.70, and 0.90 with 10 random seeds per circuit and value function. Here, a sampling fraction refers to the fraction of the available Owen context space sampled for each gate. For example, a sampling fraction of 0.70 means that approximately 70% of the exact outer–inner coalition contexts for that gate are sampled. The sampled Owen values are compared with exact Owen values using maximum absolute error, normalized error, Spearman rank correlation, and runtime. For an active-gate set N, the normalized error is defined as

$$\mathrm{NE} = \frac{\frac{1}{|N|}\sum_{i\in N}|\widehat{\psi}_i - \psi_i|}{\sum_{i\in N}|\psi_i| + 10^{-12}}. \qquad (17)$$

Here, $\widehat{\psi}_i$ is the sampled Owen value and ψ_i is the exact Owen value of gate i. Thus, the normalized error is the mean absolute difference between sampled and exact gate-level Owen values, divided by the total absolute magnitude of the exact gate-level Owen-value vector. This makes the error relative to the scale of the exact attribution vector. Based on this validation, the full 35-circuit study uses sampling fraction 0.70 and averages five sampled runs per circuit and value function.

E. Meaningfulness and Faithfulness Criteria

The benchmark evaluates both meaningfulness and behavioural faithfulness. Meaningfulness is assessed by comparing the dominant Owen groups with the expected circuit roles. The dominant group is defined as the group with the largest signed group-level Owen total Ψ_G. For the magic value function, M or X is expected to be important because these groups contain non-Clifford or magic-related gates. For the entanglement value function, E is expected to be important because it contains the main entangling gates. Interior circuits are expected to show more mixed attributions.

Faithfulness is evaluated using group ablation. For each circuit and value function, the highest-ranked and lowest-ranked Owen groups are removed separately from the full circuit, and the value function is recomputed. For a group G, the value drop is

$$\Delta_G = v(N) - v(N \setminus G). \qquad (18)$$

A faithful ranking should usually produce a larger drop for the top-ranked group than for the bottom-ranked group. This is a behavioural check rather than a proof, because near-zero values and tied group attributions can make the top–bottom comparison ambiguous.

IV. Experiments

The experiments apply the Owen-value framework to two SVQX QML classification case studies [5]: a stochastic one-shot QNN accuracy setting and a deterministic kernel-based QSVM setting. Together, they test whether Owen values add useful module-level interpretations beyond the original gate-level Shapley/SVQX explanations.

A. Quantum Neural Network Case Study

The first QML case study uses a quantum neural network (QNN). A QNN is a classifier in which a parameterized quantum circuit processes an input and a measured qubit is used to predict the class label [3]. This case study follows the two-qubit QNN experiment from SVQX [5]. The circuit and gate indexing are reported in Appendix A-C. The trained parameters are kept fixed at $\theta \approx (3.860, -1.070, -1.583, 0.860)$ during the explanation experiment, and the circuit is evaluated on the same 20-point binary classification dataset [5]. The main E/M/X coalition structure for this QNN is reported in Appendix A-C.

The value function is classification accuracy, as defined in Eq. (15). The QNN circuit is evaluated with Qiskit Aer `qasm_simulator` using one shot per input. Therefore, each prediction is obtained from a single simulated measurement outcome, making the accuracy value function stochastic. To reduce this uncertainty, each coalition value is averaged over $K \in \{1, 8, 16, 32\}$ repeated evaluations, following Section II-B. Each setting is repeated over five independent runs. The resulting Owen explanations are compared with the original gate-level SVQX/Shapley pattern.

B. Quantum Support Vector Machine Case Study

The second QML case study applies Owen values to the quantum support vector machine (QSVM) setup from SVQX [5]. A QSVM uses a quantum feature map to embed classical data into quantum states, then trains a classical support vector

TABLE I
SUMMARY CHECKS FOR EXACT OWEN VALUES ON THE 15-CIRCUIT SUBSET.

Check	Result
Magic-axis circuits with top magic group in $\{M, X\}$	5/5
Magic-axis circuits with near-zero entanglement attribution	5/5
Entanglement-axis circuits with strict top group E	3/5
Entanglement-axis circuits where E is top or co-top	5/5
Interior circuits with shared or mixed attribution	5/5

classifier on the resulting quantum kernel [18], [19]. The feature map is a two-qubit second-order Pauli-Z map with $r \in \{1, 2, 3\}$ repetitions; the gate indexing and E/M/X structure are reported in Appendix A-G. The phase functions are $\phi_1(x_1) = 2x_1$, $\phi_2(x_2) = 2x_2$, and $\phi(x) = 2(\pi - x_1)(\pi - x_2)$. The E/M/X coalition structure for each repetition block is reported in Appendix A-G.

For each coalition S, the reduced feature map defines quantum states for all data points. Kernel entries are computed as squared state overlaps using deterministic statevector simulation: the main implementation uses a manual NumPy statevector simulator, with Qiskit's `Statevector` simulator used as a reference check. A classical support vector classifier is then trained with the resulting precomputed kernel, and the coalition value is the test accuracy from Eq. (15).

Five train/test splits are used, each with 40 training points and 1000 test points. Since both the kernel evaluation and classifier training are deterministic, the QSVM value function gives the same result on repeated evaluations, so $K = 1$ is sufficient. Owen contexts are enumerated exactly for this case study.

V. RESULTS

This section reports the experimental results for the controlled benchmark, the Monte Carlo estimator validation, and the two QML case studies.

A. Exact Owen Values on the 15-Circuit Subset

The exact 15-circuit subset contains five magic-axis circuits, five entanglement-axis circuits, and five interior circuits. Table I summarizes the main group-level checks.

For the magic-axis circuits, the dominant magic group is in $\{M, X\}$ in all five cases, and the entanglement attribution is near zero in all five cases. For the entanglement-axis circuits, E is the strict top group in three out of five cases and is top or co-top in all five cases. The five interior circuits show shared or mixed attribution patterns.

B. Faithfulness by Group Ablation

Group ablation compares the value drop after removing the top-ranked Owen group with the value drop after removing the bottom-ranked Owen group. The results are shown in Fig. 2 and summarized in Table II.

For the magic value function, the mean top-group drop is 0.3502, compared with 0.0936 for the bottom group, and the top group causes the larger drop in 13 out of 15 circuits. The two non-winning cases correspond to one circuit with a single active group and one circuit with near-zero magic.

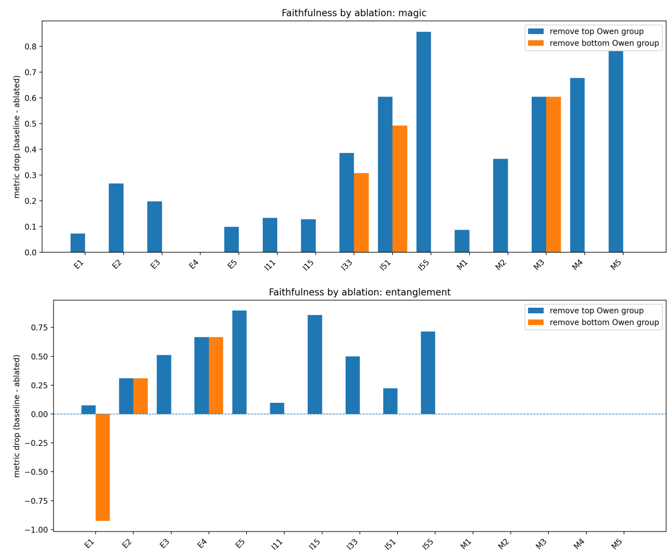

Fig. 2. Faithfulness by ablation on the exact 15-circuit subset. The bars compare the value drop after removing the top-ranked Owen group and the bottom-ranked Owen group. Magic gives the clearest signal, while entanglement contains ties and near-zero baseline cases.

TABLE II
FAITHFULNESS ABLATION SUMMARY ON THE EXACT 15-CIRCUIT SUBSET.

Value function	Mean top drop	Mean bottom drop	Top wins
Magic	0.3502	0.0936	13/15
Entanglement	0.3233	0.0032	8/15

For the entanglement value function, the mean top-group drop is 0.3233, compared with 0.0032 for the bottom group, and the top group causes the larger drop in 8 out of 15 circuits. Out of 7 non-winning cases 5 have near-zero entanglement.

C. Monte Carlo Estimator Validation

The Monte Carlo Owen estimator was validated against exact Owen values on the 15-circuit subset. Sampling fractions 0.30, 0.50, 0.70, and 0.90 were tested with 10 random seeds per circuit and value function. In Fig. 3, results are first averaged over seeds for each circuit and then averaged over the 15 circuits; shaded regions show ± 1 standard deviation across circuits.

As shown in Fig. 3, increasing the sampling fraction generally decreases error and increases runtime; the numerical values are reported in Appendix A-B. For magic, the normalized error decreases from 0.059 at 30% to 0.032 at 70% and remains 0.032 at 90%. For entanglement, the normalized error decreases from 0.061 at 30% to 0.041 at 70% and 0.037 at 90%. At 70%, the mean Spearman correlation is approximately 0.924 for both value functions. The full 35-circuit experiment therefore uses 70% sampling and averages five sampled runs per circuit and value function.

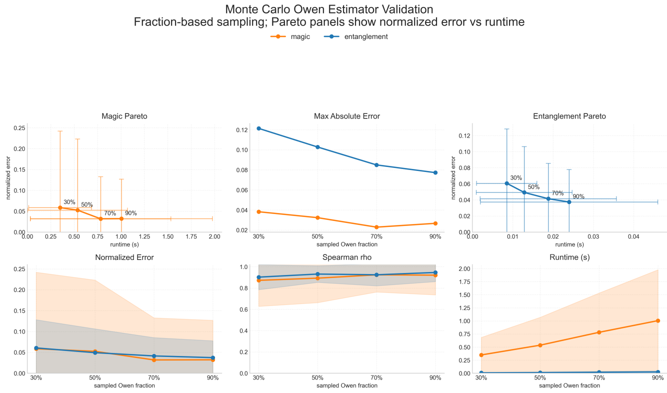

Fig. 3. Monte Carlo Owen estimator validation on the exact 15-circuit subset. Errors generally decrease as the sampling fraction increases, while runtime increases. Shaded regions show ± 1 standard deviation across circuits.

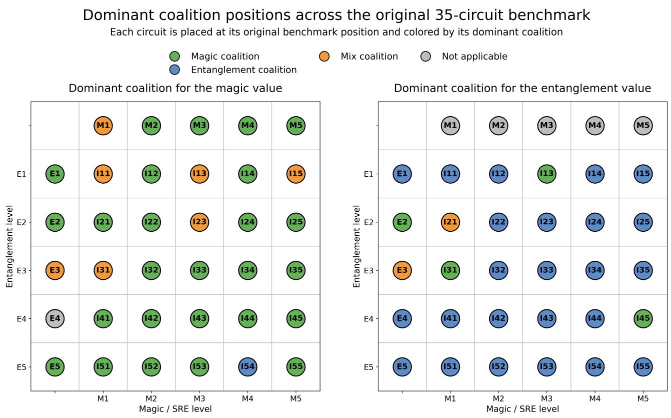

Fig. 4. Dominant Owen group on the full 35-circuit benchmark. Grey cells indicate circuits with numerically zero target value.

D. Estimated Owen Values on the Full 35-Circuit Benchmark

The full benchmark applies the sampled Owen estimator with sampling fraction 0.70 to all 35 circuits, averaging five sampled runs per circuit and value function. Figure 4 shows the dominant Owen group for each circuit. Circuits with numerically zero target value are shown in grey.

For the magic value, one circuit is not applicable because its SRE is zero. Among the remaining circuits, M is dominant in 26 circuits, X in 7 circuits, and E in 1 circuit. For the entanglement value, the five magic-axis circuits are not applicable because their entanglement value is zero. Among the remaining 30 circuits, E is dominant in 24 circuits, M in 4 circuits, and X in 2 circuits.

E. QNN Case Study

The QNN case study uses the two-qubit QNN described in Appendix A-C. Because the value function is one-shot classification accuracy, coalition values are stochastic. The E/M/X partition was evaluated with $K \in \{1, 8, 16, 32\}$ repeated value-function evaluations, and $K = 32$ is used for the final reported results. The full K comparison is reported in Appendix A-D.

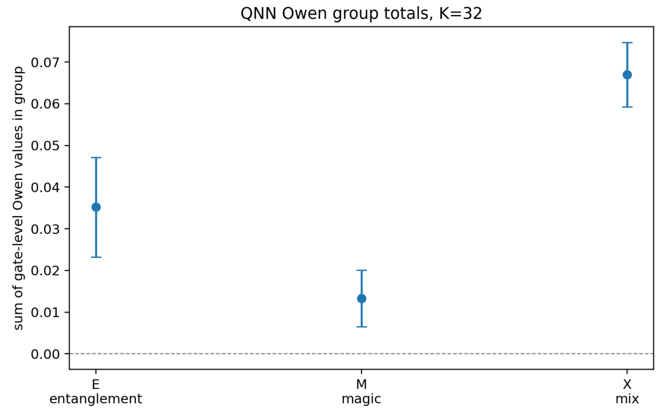

Fig. 5. Group-level QNN Owen values for the E/M/X partition with $K = 32$. The mixed group X receives the largest total attribution.

For $K = 32$, the largest individual gate-level Owen value is assigned to gate 15 in the mixed group X. Additional positive contributions occur around gates 7, 16, and 17. The full gate-level plot is reported in Appendix A-E.

The group-level totals in Fig. 5 are

$$E = 0.0352, \qquad M = 0.0133, \qquad X = 0.0670.$$

Thus, X has the largest group-level Owen total, followed by E and M.

Additional QNN partitions are reported in Appendix A-F.

F. QSVM Case Study

The QSVM case study applies the E/M/X Owen explanation to the SVQX feature-map setup described in Appendix A-G. The value function is the QSVM test accuracy after retraining the classifier on the coalition kernel. Hadamard gates are treated as passive, while active gates are grouped into CNOT scaffold gates E, local phase rotations M, and cross-feature motif gates X; exact gate indices are listed in Appendix A-G. Full-circuit accuracy checks are reported in Appendix A-H.

The mean gate-level Owen values over five datasets have their largest values on phase-encoding gates and intermediate feature-map gates, while final CNOT gates are often close to zero. The full gate-level plot is reported in Appendix A-I.

The group-level Owen values in Fig. 6 show that M has the largest total attribution for all three values of r. The mixed group X is larger for $r = 2$ and $r = 3$ than for $r = 1$, while E remains smaller than M in all three settings.

VI. Discussion

A. Interpretation of the Results

The benchmark results show that Owen values are sensitive to the value function being explained. On the exact 15-circuit subset, magic-axis circuits are assigned mainly to M or X, while entanglement-axis circuits assign E as top or co-top. The full 35-circuit benchmark shows the same overall pattern at larger scale: magic is mostly M-dominated, while entanglement is mostly E-dominated. This supports the idea

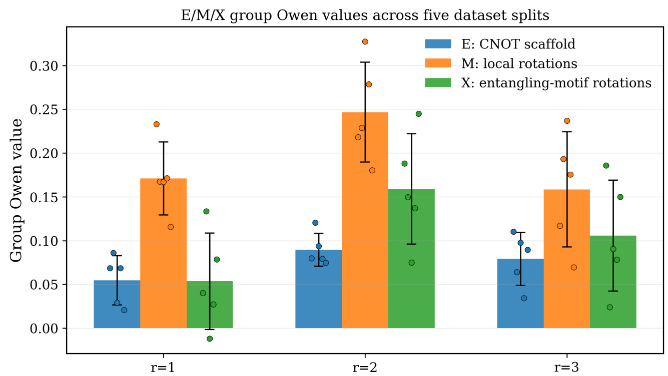

Fig. 6. QSVM group-level Owen values for the E/M/X partition. The local rotation group M is largest, while the mixed group X becomes more important for $r = 2$ and $r = 3$.

that the attributions are not merely reflecting the manually chosen E/M/X labels, but the reduced-circuit game induced by each value function.

The mixed and exceptional cases are also important. Some magic explanations involve X, and some entanglement explanations involve M or X. This shows that quantum resources are not always cleanly localized in one type of gate: local rotations can affect the states on which entangling gates act, and mixed gates can mediate local–entangling effects. This is consistent with the broader view that entanglement and magic can interact in circuit dynamics [8].

The ablation experiment gives behavioural support to the Owen rankings, but also shows the limits of such a test. The faithfulness signal is clearer for magic than for entanglement. For entanglement, zero target values, tied attributions, and degenerate active groups make the top–bottom comparison less decisive. Overall, the ablation results support behavioural faithfulness, especially for magic, while showing that faithfulness checks are less informative when the target value is close to zero or when several groups interact.

The Monte Carlo validation shows that sampled Owen values can preserve the main ranking structure while reducing exact enumeration cost. The chosen sampling fraction gives a practical trade-off between runtime and accuracy for the full benchmark. However, this validation is still performed on small circuits where exact Owen values are available, so larger circuits may require additional estimator checks.

The QNN and QSVM case studies show that Owen values can add a module-level view to SVQX-style gate-level explanations. In the QNN case study, repeated coalition evaluations reduce the instability caused by the one-shot accuracy value function, and the largest group-level contribution comes from the mixed local–entangling group. In the QSVM case study, the local phase-rotation group dominates across repetition depths, while the mixed group becomes more important for deeper feature maps. In both cases, the gate-level patterns remain compatible with the original Shapley/SVQX explanations [5], while the group totals provide a more compact interpretation of which circuit roles matter. This is useful because quantum circuits are often designed as structured blocks rather than as unrelated individual gates [7].

B. Limitations and Future Work

The main limitation is the dependence on the chosen coalition structure. A meaningful partition can reveal useful module-level behaviour, whereas a poor partition may give weak or misleading explanations. Owen explanations should therefore always report the exact gate grouping and its motivation. The experiments are also limited to small three-qubit benchmark circuits and controlled SVQX case studies, so larger circuits, noisy simulations, hardware runs, and more complex QML models may behave differently.

Future work should study automatic or data-driven coalition construction, evaluate the method under realistic noise and hardware conditions, and test whether Owen values can support quantum architecture search by identifying useful functional modules.

VII. Conclusion

This thesis investigated whether Owen values can provide group-structured explanations for quantum circuits. For RQ1, the thesis showed that Owen values can be adapted to SVQX-style explanations by treating active gates as players, predefined circuit modules as groups, and reduced circuits as coalition evaluations. The same framework can explain different value functions, including magic, entanglement, and classification accuracy, and can be computed exactly or approximated with Monte Carlo sampling.

For RQ2, the controlled benchmark showed that Owen values produce meaningful group-wise explanations. Magic was mostly attributed to local or mixed non-Clifford structure, while entanglement was mostly attributed to entangling gates. Group ablation provided behavioural support for these rankings, especially for magic, while also showing that near-zero values, ties, and group interactions can make faithfulness harder to assess.

For RQ3, the QNN and QSVM case studies showed that Owen values preserve useful gate-level SVQX/Shapley patterns while adding a module-level interpretation. Thus, Owen values do not replace Shapley values, but answer a complementary question: which predefined functional circuit groups matter for the chosen value function?

The main limitation is the dependence on the chosen coalition structure. Future work should study automatic group construction, larger and noisier circuits, hardware-based evaluation, and possible use in quantum architecture search. More interpretable quantum-circuit explanations can also have broader societal value: as quantum machine learning develops, methods that make model behaviour easier to inspect can support transparency, debugging, and more responsible use of quantum technologies.

Acknowledgment

I would like to thank my supervisors, Menica Dibenedetto and Tjitze Rienstra, for their guidance, feedback, and support throughout this thesis. I am also grateful to the authors of the SVQX and SRE-estimation repositories, whose publicly available code and papers helped shape the experimental basis of this work. Finally, I thank the Department of Advanced Computing Sciences at Maastricht University for providing the academic environment in which this thesis was developed.

Appendix A
Coalition Structures and Additional Experiment Results

This appendix reports the exact gate partitions used in the controlled 35-circuit benchmark, QNN case study, and QSVM case study, together with additional stability checks and supporting result tables. All gate indices are thesis-style 1-based indices.

A. Controlled Benchmark Coalition Structures

Tables III–IV list the gate sequences of the controlled benchmark circuits. Each entry has the form *index:gate(qubit)*. For CNOT gates, $q_a \to q_b$ denotes control qubit q_a and target qubit q_b.

Tables V–VI report the corresponding E/M/X coalition assignments using 1-based gate indices. Passive gates are kept in every reduced circuit but do not receive Owen attributions.

A dash indicates that no gate in that circuit belongs to the corresponding role; empty roles are omitted from the active partition during Owen-value computation.

B. Estimator Validation Results

Table VII reports the numerical results used to validate the Monte Carlo Owen estimator on the exact 15-circuit subset.

C. QNN Coalition Structures

Figure 7 shows the two-qubit QNN circuit and the gate indexing used in the Owen-value analysis. The main QNN result in Section V-E uses the resource-based E/M/X partition shown in Table VIII.

D. QNN Stability Across Repeated Evaluations

The main QNN result uses $K = 32$ repeated value-function evaluations per coalition. Figure 8 shows the corresponding stability check for $K \in \{1, 8, 16, 32\}$. The estimates are noisier for small K, especially for gates with small contributions, while the main attribution pattern remains stable as K increases.

E. QNN Gate-Level Owen Results

Figure 9 reports the final gate-level QNN Owen values for the E/M/X partition with $K = 32$.

F. Additional QNN Partitions

In addition to the resource-based E/M/X partition, three alternative QNN partitions were tested as a sensitivity analysis. These partitions are not the main focus of the QNN case study, but they show that Owen values can answer different module-level questions depending on the chosen coalition structure. For these partitions, gates $\{1, 3\}$ are passive, following the original SVQX QNN setup.

Table X reports the group-level Owen summaries for these additional partitions. The architecture partition assigns the largest total attribution to the trained classifier block C. The feature-interaction partition highlights local feature encoding L and the classifier C, while the explicit cross-feature entangling motif group XFE is close to zero. The qubit-role partition assigns the largest total attribution to the measured-qubit path $Q0$, which is consistent with the fact that the model prediction is read from the first qubit.

G. QSVM Coalition Structure

Figure 10 shows one repetition block of the QSVM feature map and the gate indexing used in the Owen-value analysis. The QSVM feature map consists of $r \in \{1, 2, 3\}$ repetitions of a seven-gate block. For repetition $j \in \{1, \ldots, r\}$, the gate indices in that block are shifted by $7(j - 1)$. The coalition structure is listed in Table XI. The Hadamard gates are passive, the CNOT gates form the entanglement group, the local phase gates form the magic group, and the cross-feature phase gate forms the mixed group.

H. Additional QSVM Results

Table XII reports the full-circuit QSVM accuracy over five train/test splits. This is used as a setup check before interpreting the Owen attributions.

Table XIII reports the group-level Owen values.

I. QSVM Gate-Level Owen Results

Figure 11 reports the mean gate-level QSVM Owen values over five datasets for $r \in \{1, 2, 3\}$.

TABLE III

GATE DEFINITIONS FOR THE MAGIC-AXIS AND ENTANGLEMENT-AXIS BENCHMARK CIRCUITS.

Circuit	Gate sequence
M1	1 : ry(q_0), 2 : h(q_0), 3 : s(q_1), 4 : t(q_1), 5 : cx($q_1 \rightarrow q_0$), 6 : rx(q_1), 7 : rx(q_0), 8 : rz(q_1)
M2	1 : rz(q_0), 2 : rz(q_1), 3 : cx($q_1 \rightarrow q_2$), 4 : rz(q_1), 5 : rx(q_0), 6 : rx(q_2), 7 : rz(q_0), 8 : rz(q_1)
M3	1 : rx(q_0), 2 : rx(q_1), 3 : h(q_1), 4 : rz(q_0), 5 : rz(q_1), 6 : rx(q_2)
M4	1 : cx($q_0 \rightarrow q_1$), 2 : rx(q_2), 3 : ry(q_1), 4 : rx(q_1), 5 : rx(q_1), 6 : cx($q_0 \rightarrow q_2$), 7 : ry(q_0), 8 : rz(q_2)
M5	1 : ry(q_0), 2 : rz(q_1), 3 : cx($q_2 \rightarrow q_1$), 4 : rx(q_1), 5 : rz(q_0), 6 : rx(q_2), 7 : rz(q_2)
E1	1 : rx(q_0), 2 : rz(q_0), 3 : rx(q_0), 4 : s(q_0), 5 : h(q_1), 6 : cx($q_1 \rightarrow q_0$)
E2	1 : rz(q_0), 2 : rx(q_1), 3 : rx(q_1), 4 : ry(q_2), 5 : cx($q_1 \rightarrow q_0$), 6 : cx($q_1 \rightarrow q_2$)
E3	1 : rx(q_0), 2 : cx($q_1 \rightarrow q_0$), 3 : rz(q_2), 4 : s(q_2), 5 : t(q_2), 6 : rz(q_2), 7 : cx($q_0 \rightarrow q_1$)
E4	1 : h(q_0), 2 : h(q_0), 3 : z(q_0), 4 : h(q_1), 5 : x(q_1), 6 : s(q_1), 7 : cx($q_1 \rightarrow q_2$), 8 : h(q_0)
E5	1 : ry(q_0), 2 : t(q_1), 3 : rx(q_2), 4 : cx($q_0 \rightarrow q_2$), 5 : rz(q_1), 6 : cx($q_2 \rightarrow q_1$)

TABLE IV

GATE DEFINITIONS FOR THE INTERIOR BENCHMARK CIRCUITS.

Circuit	Gate sequence
I11	1 : t(q_0), 2 : rz(q_1), 3 : ry(q_2), 4 : cx($q_0 \rightarrow q_2$), 5 : cx($q_2 \rightarrow q_1$), 6 : h(q_1)
I12	1 : ry(q_0), 2 : rz(q_1), 3 : cx($q_0 \rightarrow q_1$), 4 : t(q_1), 5 : t(q_0), 6 : h(q_0), 7 : h(q_0), 8 : h(q_1), 9 : s(q_0), 10 : cx($q_1 \rightarrow q_0$), 11 : cx($q_0 \rightarrow q_1$)
I13	1 : s(q_0), 2 : h(q_0), 3 : t(q_1), 4 : rz(q_2), 5 : cx($q_1 \rightarrow q_0$), 6 : h(q_1), 7 : h(q_0), 8 : rx(q_1), 9 : s(q_0), 10 : t(q_2), 11 : rz(q_0), 12 : rz(q_0), 13 : rz(q_1), 14 : ry(q_2), 15 : cx($q_2 \rightarrow q_1$)
I14	1 : rx(q_0), 2 : ry(q_0), 3 : rz(q_1), 4 : cx($q_1 \rightarrow q_2$), 5 : rx(q_0), 6 : ry(q_0), 7 : rx(q_2), 8 : cx($q_0 \rightarrow q_2$), 9 : cx($q_2 \rightarrow q_1$), 10 : ry(q_1), 11 : cx($q_0 \rightarrow q_2$), 12 : rz(q_1), 13 : ry(q_2)
I15	1 : rz(q_0), 2 : ry(q_0), 3 : rx(q_1), 4 : cx($q_2 \rightarrow q_1$), 5 : cx($q_1 \rightarrow q_0$), 6 : cx($q_0 \rightarrow q_2$)
I21	1 : ry(q_0), 2 : cx($q_1 \rightarrow q_0$), 3 : cx($q_0 \rightarrow q_2$), 4 : rz(q_1), 5 : rz(q_0), 6 : rz(q_2), 7 : rx(q_0), 8 : cx($q_1 \rightarrow q_2$), 9 : ry(q_0), 10 : ry(q_1), 11 : ry(q_0), 12 : rx(q_0), 13 : ry(q_0), 14 : cx($q_2 \rightarrow q_0$)
I22	1 : rx(q_0), 2 : cx($q_1 \rightarrow q_2$), 3 : cx($q_1 \rightarrow q_2$), 4 : rz(q_1), 5 : rz(q_0), 6 : rz(q_0), 7 : rx(q_0), 8 : rx(q_2), 9 : rx(q_0), 10 : cx($q_2 \rightarrow q_0$)
I23	1 : z(q_0), 2 : t(q_1), 3 : h(q_2), 4 : t(q_0), 5 : h(q_0), 6 : cx($q_2 \rightarrow q_0$), 7 : t(q_0), 8 : x(q_0), 9 : h(q_0), 10 : t(q_0), 11 : y(q_2), 12 : cx($q_2 \rightarrow q_0$), 13 : s(q_2)
I24	1 : rx(q_0), 2 : rx(q_1), 3 : cx($q_2 \rightarrow q_0$), 4 : cx($q_0 \rightarrow q_2$), 5 : ry(q_1), 6 : h(q_2), 7 : ry(q_1), 8 : t(q_1), 9 : h(q_2), 10 : h(q_0), 11 : cx($q_1 \rightarrow q_2$), 12 : rz(q_1), 13 : s(q_1), 14 : h(q_0), 15 : rx(q_1)
I25	1 : s(q_0), 2 : h(q_0), 3 : cx($q_1 \rightarrow q_2$), 4 : rz(q_0), 5 : t(q_1), 6 : cx($q_0 \rightarrow q_1$), 7 : rx(q_2), 8 : rz(q_0), 9 : t(q_1), 10 : h(q_1), 11 : ry(q_2), 12 : s(q_1), 13 : h(q_1), 14 : rx(q_2), 15 : h(q_1), 16 : cx($q_1 \rightarrow q_2$), 17 : h(q_1)
I31	1 : ry(q_0), 2 : rz(q_1), 3 : rz(q_2), 4 : rx(q_2), 5 : cx($q_0 \rightarrow q_1$), 6 : rx(q_0), 7 : ry(q_0), 8 : rz(q_2), 9 : rx(q_2), 10 : cx($q_2 \rightarrow q_1$), 11 : rx(q_1), 12 : rz(q_0), 13 : cx($q_2 \rightarrow q_1$), 14 : rz(q_0), 15 : cx($q_2 \rightarrow q_1$)
I32	1 : rz(q_0), 2 : cx($q_1 \rightarrow q_0$), 3 : cx($q_1 \rightarrow q_2$), 4 : s(q_1), 5 : h(q_1), 6 : h(q_2), 7 : rz(q_2), 8 : rz(q_1), 9 : s(q_0), 10 : cx($q_1 \rightarrow q_2$), 11 : ry(q_2), 12 : rz(q_2), 13 : s(q_0), 14 : s(q_1), 15 : cx($q_2 \rightarrow q_0$), 16 : h(q_1), 17 : cx($q_2 \rightarrow q_0$), 18 : rx(q_1)
I33	1 : rx(q_0), 2 : cx($q_0 \rightarrow q_1$), 3 : ry(q_0), 4 : cx($q_0 \rightarrow q_1$), 5 : rz(q_2), 6 : ry(q_0), 7 : rz(q_1)
I34	1 : cx($q_0 \rightarrow q_1$), 2 : cx($q_2 \rightarrow q_1$), 3 : ry(q_2), 4 : ry(q_2), 5 : ry(q_2), 6 : rx(q_1), 7 : rx(q_2), 8 : cx($q_1 \rightarrow q_0$), 9 : cx($q_1 \rightarrow q_2$), 10 : rx(q_0), 11 : rz(q_1), 12 : rz(q_1), 13 : cx($q_1 \rightarrow q_0$), 14 : cx($q_2 \rightarrow q_0$)
I35	1 : cx($q_0 \rightarrow q_1$), 2 : rz(q_0), 3 : rz(q_0), 4 : ry(q_1), 5 : ry(q_1), 6 : rx(q_1), 7 : rz(q_1), 8 : rz(q_1), 9 : rz(q_0), 10 : rx(q_0), 11 : cx($q_1 \rightarrow q_2$), 12 : ry(q_1), 13 : cx($q_1 \rightarrow q_0$), 14 : ry(q_2), 15 : ry(q_2)
I41	1 : s(q_0), 2 : t(q_1), 3 : h(q_0), 4 : ry(q_0), 5 : ry(q_2), 6 : rz(q_2), 7 : rx(q_0), 8 : h(q_1), 9 : rx(q_2), 10 : rz(q_0), 11 : rx(q_2), 12 : ry(q_0), 13 : s(q_1), 14 : cx($q_0 \rightarrow q_2$), 15 : rx(q_2), 16 : h(q_0), 17 : rx(q_2), 18 : t(q_0), 19 : rx(q_1)
I42	1 : rx(q_0), 2 : cx($q_1 \rightarrow q_2$), 3 : rx(q_1), 4 : cx($q_0 \rightarrow q_1$), 5 : ry(q_1), 6 : rz(q_1), 7 : ry(q_0), 8 : ry(q_2), 9 : rz(q_2), 10 : rz(q_2), 11 : rz(q_0), 12 : rz(q_2), 13 : rz(q_1)
I43	1 : rx(q_0), 2 : cx($q_1 \rightarrow q_0$), 3 : ry(q_2), 4 : rz(q_2), 5 : ry(q_2), 6 : rz(q_1), 7 : rx(q_1), 8 : cx($q_2 \rightarrow q_1$), 9 : rz(q_0), 10 : ry(q_1), 11 : rx(q_0), 12 : rx(q_2), 13 : ry(q_0), 14 : cx($q_0 \rightarrow q_1$)
I44	1 : ry(q_0), 2 : cx($q_1 \rightarrow q_2$), 3 : rz(q_0), 4 : ry(q_2), 5 : rx(q_1), 6 : cx($q_1 \rightarrow q_0$), 7 : cx($q_0 \rightarrow q_1$), 8 : rz(q_2), 9 : cx($q_2 \rightarrow q_1$), 10 : ry(q_2), 11 : rz(q_1), 12 : ry(q_2), 13 : rx(q_1), 14 : rz(q_0)
I45	1 : s(q_0), 2 : h(q_1), 3 : s(q_2), 4 : h(q_0), 5 : cx($q_1 \rightarrow q_2$), 6 : h(q_2), 7 : rx(q_2), 8 : rz(q_0), 9 : ry(q_1), 10 : s(q_2), 11 : cx($q_2 \rightarrow q_0$), 12 : h(q_1), 13 : rx(q_0), 14 : cx($q_1 \rightarrow q_2$)
I51	1 : rx(q_0), 2 : rz(q_0), 3 : ry(q_1), 4 : rx(q_2), 5 : cx($q_1 \rightarrow q_2$), 6 : ry(q_1), 7 : ry(q_2)
I52	1 : cx($q_0 \rightarrow q_1$), 2 : rx(q_0), 3 : rx(q_2), 4 : cx($q_0 \rightarrow q_2$), 5 : cx($q_0 \rightarrow q_2$), 6 : ry(q_1), 7 : ry(q_2), 8 : ry(q_0), 9 : cx($q_1 \rightarrow q_0$), 10 : rx(q_0), 11 : rz(q_1), 12 : rx(q_1), 13 : rz(q_0), 14 : rx(q_0), 15 : cx($q_0 \rightarrow q_1$)
I53	1 : h(q_0), 2 : rx(q_0), 3 : s(q_1), 4 : cx($q_1 \rightarrow q_2$), 5 : s(q_2), 6 : cx($q_0 \rightarrow q_1$), 7 : ry(q_0), 8 : cx($q_0 \rightarrow q_1$), 9 : h(q_1), 10 : ry(q_2), 11 : rz(q_2), 12 : rx(q_2), 13 : t(q_0), 14 : cx($q_2 \rightarrow q_1$), 15 : rx(q_1), 16 : h(q_0), 17 : s(q_0), 18 : rx(q_0), 19 : rz(q_1)
I54	1 : ry(q_0), 2 : cx($q_0 \rightarrow q_1$), 3 : cx($q_2 \rightarrow q_0$), 4 : rx(q_0), 5 : cx($q_1 \rightarrow q_0$), 6 : rz(q_0), 7 : rz(q_2), 8 : rz(q_1), 9 : ry(q_2), 10 : rz(q_0), 11 : ry(q_2), 12 : cx($q_1 \rightarrow q_0$), 13 : cx($q_2 \rightarrow q_0$), 14 : ry(q_0), 15 : rx(q_0), 16 : cx($q_1 \rightarrow q_0$), 17 : ry(q_2), 18 : rz(q_0), 19 : rz(q_1)
I55	1 : rx(q_0), 2 : ry(q_1), 3 : rz(q_1), 4 : rx(q_2), 5 : rz(q_1), 6 : cx($q_0 \rightarrow q_1$), 7 : cx($q_1 \rightarrow q_0$), 8 : cx($q_2 \rightarrow q_0$)

TABLE V

E/M/X COALITION STRUCTURES FOR THE MAGIC-AXIS AND ENTANGLEMENT-AXIS BENCHMARK CIRCUITS.

Circuit	Passive	E	M	X
M1	{2, 3}	{5}	{1, 7, 8}	{4, 6}
M2	–	{3}	{1, 5, 6, 7, 8}	{2, 4}
M3	{3}	–	{1, 2, 4, 5, 6}	–
M4	–	{1, 6}	{2, 3, 4, 5, 8}	{7}
M5	–	{3}	{1, 5, 6, 7}	{2, 4}
E1	{4, 5}	{6}	{1, 2, 3}	–
E2	–	{5, 6}	{1, 2, 3, 4}	–
E3	{4}	{2, 7}	{3, 5, 6}	{1}
E4	{1, 2, 3, 4, 5, 6, 8}	{7}	–	–
E5	–	{4, 6}	{1, 2}	{3, 5}

TABLE VI
E/M/X COALITION STRUCTURES FOR INTERIOR BENCHMARK CIRCUITS.

Circuit	Passive	E	M	X
I11	{6}	{4, 5}	{1, 2}	{3}
I12	{6, 7, 8, 9}	{3, 10, 11}	{1, 5}	{2, 4}
I13	{1, 2, 6, 7, 9}	{5, 15}	{3, 4, 8, 10, 11, 12, 13}	{14}
I14	–	{4, 8, 9, 11}	{1, 2, 5, 6, 12, 13}	{3, 7, 10}
I15	–	{4, 5, 6}	{1, 2}	{3}
I21	–	{2, 3, 8, 14}	{4, 5, 6, 7, 9, 10, 11, 12}	{1, 13}
I22	–	{2, 3, 10}	{1, 5, 6, 7, 8}	{4, 9}
I23	{1, 3, 5, 8, 9, 11, 13}	{6, 12}	{2, 4, 10}	{7}
I24	{6, 9, 10, 13, 14}	{3, 4, 11}	{1, 2, 5, 7, 8, 15}	{12}
I25	{1, 2, 10, 12, 13, 15, 17}	{3, 6, 16}	{4, 7, 8, 9, 11, 14}	{5}
I31	–	{5, 10, 13, 15}	{1, 2, 3, 4, 7, 8, 12, 14}	{6, 9, 11}
I32	{4, 5, 6, 9, 13, 14, 16}	{2, 3, 10, 15, 17}	{7, 8, 12, 18}	{1, 11}
I33	–	{2, 4}	{5, 6, 7}	{1, 3}
I34	–	{1, 2, 8, 9, 13, 14}	{4, 5, 6, 7, 10, 11}	{3, 12}
I35	–	{1, 11, 13}	{3, 4, 5, 6, 7, 8, 9, 10, 14, 15}	{2, 12}
I41	{1, 3, 8, 13, 16}	{14}	{2, 4, 5, 6, 7, 9, 10, 11, 12, 17, 18, 19}	{15}
I42	–	{2, 4}	{1, 6, 7, 8, 9, 10, 11, 12, 13}	{3, 5}
I43	–	{2, 8, 14}	{3, 4, 5, 6, 9, 10, 11, 12}	{1, 7, 13}
I44	–	{2, 6, 7, 9}	{1, 3, 4, 11, 12, 13, 14}	{5, 8, 10}
I45	{1, 2, 3, 4, 6, 10, 12}	{5, 11, 14}	{7, 8, 9, 13}	–
I51	–	{5}	{1, 2, 3, 7}	{4, 6}
I52	–	{1, 4, 5, 9, 15}	{6, 7, 11, 12, 13}	{2, 3, 8, 10, 14}
I53	{1, 3, 5, 9, 16, 17}	{4, 6, 8, 14}	{2, 10, 11, 12, 13, 18, 19}	{7, 15}
I54	–	{2, 3, 5, 12, 13, 16}	{7, 8, 9, 10, 11, 17, 18, 19}	{1, 4, 6, 14, 15}
I55	–	{6, 7, 8}	{1, 2, 3, 4}	{5}

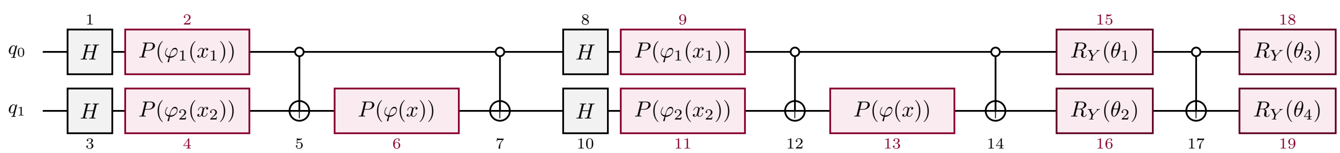

Fig. 7. Two-qubit QNN circuit used in the case study, redrawn from the SVQX experiment in [5]. Gate numbers correspond to the indexing used in the Owen-value analysis. The trained parameters are fixed at $\theta \approx (3.860, -1.070, -1.583, 0.860)$ during the explanation experiment.

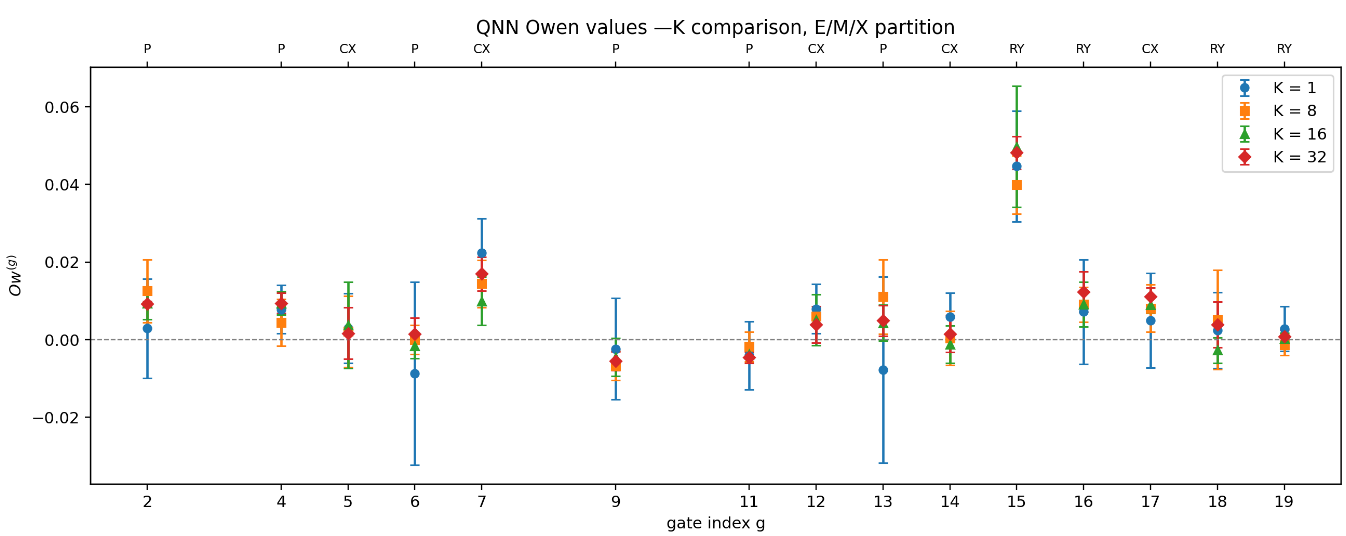

Fig. 8. Gate-level QNN Owen values under the E/M/X partition for different numbers of repeated value-function evaluations K. Error bars show variation over five independent runs.

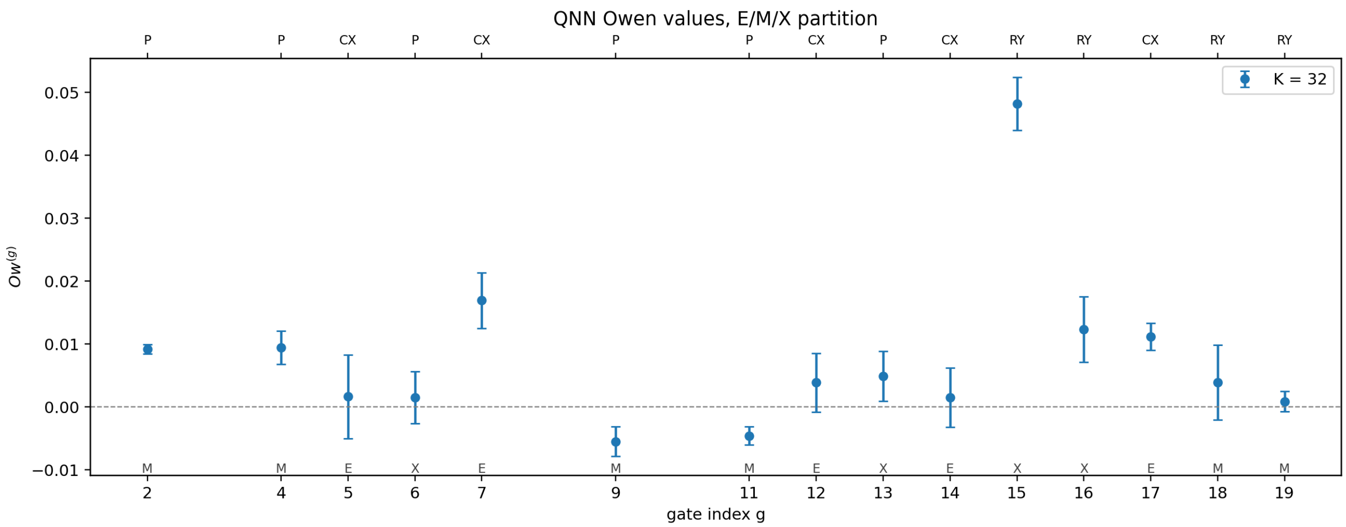

Fig. 9. Final gate-level QNN Owen values for the E/M/X partition with $K = 32$, averaged over five independent runs. The largest attribution is assigned to gate 15 in the mixed group X.

TABLE VII
ESTIMATOR-VALIDATION RESULTS. RUNTIME IS THE MEAN WALL-CLOCK TIME PER SAMPLED OWEN RUN.

Property	Fraction	Norm. error	Spearman	Runtime
Magic	0.30	0.059	0.872	0.35s
Magic	0.50	0.053	0.893	0.54s
Magic	0.70	0.032	0.924	0.78s
Magic	0.90	0.032	0.921	1.01s
Ent.	0.30	0.061	0.902	0.009s
Ent.	0.50	0.049	0.931	0.013s
Ent.	0.70	0.041	0.924	0.019s
Ent.	0.90	0.037	0.946	0.024s

TABLE VIII
RESOURCE-BASED E/M/X PARTITION FOR THE QNN CASE STUDY.

Role	Gate indices
Passive	$\{1, 3, 8, 10\}$
E	$\{5, 7, 12, 14, 17\}$
M	$\{2, 4, 9, 11, 18, 19\}$
X	$\{6, 13, 15, 16\}$

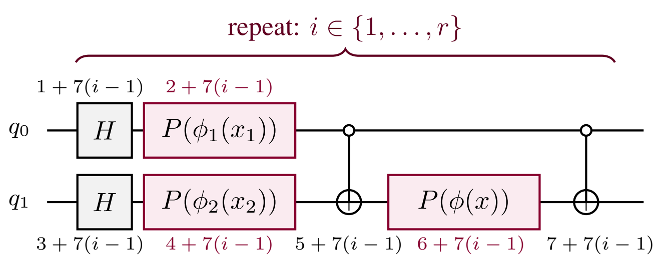

Fig. 10. One repetition block of the two-qubit QSVM feature map, redrawn from the SVQX experiment in [5]. The full feature map repeats this block $r \in \{1, 2, 3\}$ times, and gate numbers correspond to the Owen-value analysis.

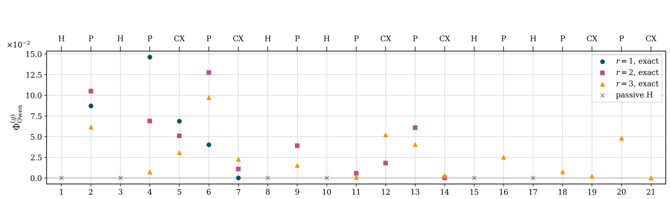

Fig. 11. Mean gate-level QSVM Owen values over five datasets for $r \in \{1, 2, 3\}$. High values repeatedly appear on phase-encoding gates and intermediate feature-map gates.

TABLE IX

ADDITIONAL QNN COALITION STRUCTURES. GATES $\{1, 3\}$ ARE PASSIVE.

Partition	Group	Gate indices
Architecture	B_1 feature-map block 1	$\{2, 4, 5, 6, 7\}$
	B_2 feature-map block 2	$\{8, 9, 10, 11, 12, 13, 14\}$
	C trainable classifier	$\{15, 16, 17, 18, 19\}$
Feature interaction	L local feature encoding	$\{2, 4, 8, 9, 10, 11\}$
	XFE cross-feature entangling motifs	$\{5, 6, 7, 12, 13, 14\}$
	C trainable classifier	$\{15, 16, 17, 18, 19\}$
Qubit role	$Q0$ measured-qubit path	$\{2, 8, 9, 15, 18\}$
	$Q1$ hidden-qubit path	$\{4, 6, 10, 11, 13, 16\}$
	$2Q$ two-qubit couplers	$\{5, 7, 12, 14, 17\}$
	$Q1F$ final hidden-qubit rotation	$\{19\}$

TABLE X

ADDITIONAL QNN OWEN SUMMARIES FOR ALTERNATIVE PARTITIONS. RESULTS USE $K = 32$ AND ARE AVERAGED OVER FIVE INDEPENDENT RUNS.

Partition	Group totals	Main interpretation
Architecture	$B_1 = 0.0323$, $B_2 = 0.0281$, $C = 0.0432$	Classifier block is largest.
Feature interaction	$L = 0.0627$, $\text{XFE} = 0.0013$, $C = 0.0479$	Local encoding and classifier dominate.
Qubit role	$Q0 = 0.0594$, $Q1 = 0.0321$, $2Q = 0.0310$, $Q1F = -0.0011$	Measured-qubit path is largest.

TABLE XI

QSVM E/M/X PARTITION FOR REPETITION j.

Role	Gate indices in repetition j
Passive	$\{1 + 7(j-1),\ 3 + 7(j-1)\}$
E	$\{5 + 7(j-1),\ 7 + 7(j-1)\}$
M	$\{2 + 7(j-1),\ 4 + 7(j-1)\}$
X	$\{6 + 7(j-1)\}$

TABLE XII

QSVM FULL-CIRCUIT ACCURACY OVER FIVE TRAIN/TEST SPLITS.

Repetitions	Mean	Std.	Range
$r = 1$	0.779	0.085	0.684–0.887
$r = 2$	0.995	0.006	0.986–1.000
$r = 3$	0.843	0.070	0.781–0.925

TABLE XIII

MEAN QSVM GROUP-LEVEL OWEN VALUES OVER FIVE DATASETS. VALUES ARE SHOWN AS MEAN $\pm$ ONE STANDARD DEVIATION.

Repetitions	E	M	X
$r = 1$	0.0546 ± 0.0282	0.1709 ± 0.0417	0.0535 ± 0.0552
$r = 2$	0.0896 ± 0.0188	0.2467 ± 0.0571	0.1590 ± 0.0630
$r = 3$	0.0791 ± 0.0302	0.1584 ± 0.0658	0.1057 ± 0.0634

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